



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS



LAB REPORT

On

Autonomus Dropping from Quadcopter using Pixhawk 4

SUBMITTED BY:

Bishal Marasini
Chanakya Aryal
Prakash Poudel
Sandip Bhattarai
Vimochan Kulung Rai
Yashaswi Shrestha

SUBMITTED TO:

Asst. Prof. Biman Rimal
Department of Mechanical &
Aerospace Engineering

Introduction to Quadcopters

Quadcopters are unmanned aerial vehicles (UAVs) with four rotors arranged in a square configuration, each driven by its own motor. These versatile aircraft have gained significant popularity in both recreational and professional applications due to their maneuverability, stability, and ability to hover in place. Unlike traditional helicopters with complex mechanical linkages, quadcopters use electronic speed controllers (ESCs) to vary the speed of each motor independently, allowing for precise control of movement.

The fundamental design of quadcopters enables six degrees of freedom: three translational movements (forward/backward, left/right, up/down) and three rotational movements (roll, pitch, and yaw). This versatility makes them ideal platforms for tasks requiring precise positioning and movement in three-dimensional space.

Working Mechanism of Quadcopters

Quadcopters operate on the principle of differential thrust. The four rotors are arranged such that diagonally opposite pairs rotate in the same direction, with one pair rotating clockwise and the other counterclockwise. This configuration balances the torque generated by the spinning propellers, allowing the quadcopter to maintain orientation.

Flight control is achieved through these key principles:

1. **Throttle:** When all motors increase or decrease speed equally, the quadcopter moves vertically up or down.
2. **Roll:** Increasing the speed of rotors on one side while decreasing the speed on the opposite side causes the quadcopter to roll, resulting in horizontal movement.
3. **Pitch:** Similar to roll, but with the front and rear pairs of motors, causing the quadcopter to pitch forward or backward, resulting in horizontal movement in those directions.
4. **Yaw:** Increasing the speed of the clockwise-rotating motors while decreasing the counterclockwise-rotating motors (or vice versa) creates a net torque, causing the quadcopter to rotate about its vertical axis.

Autonomous Dropping using Quadcopters

Autonomous dropping systems allow quadcopters to deliver payloads at specific locations without direct pilot intervention. It involves programming a quadcopter to release a payload (e.g., a package, sensor, or emergency supplies) at a specific location which is achieved by integrating a payload release mechanism with the flight controller and using mission planning software to define the drop location. This capability is valuable for applications like precision agriculture (seed or fertilizer dropping), delivery services, and emergency supply delivery.

Objectives of mission:

- To be familiar with Qground interface.
- To set up flight controller such as pixhawk for according to our requirements.
- Full autonomous operation from takeoff to landing.
- Accurate delivery to the target location.
- Optimize flight path to maximize battery life and delivery range.

Components of a Quadcopter used for Autonomous Delivery Mission

A quadcopter system consists of several essential components:

1. Frame

The structural skeleton of the quadcopter, typically made from materials like carbon fiber, aluminum, or reinforced plastic. The frame design affects the quadcopter's weight, durability, and aerodynamic properties. The X configuration S500 quadcopter frame with PCB Central Plate was used for the mission because of its balance of maneuverability and stability.

2. Motors

Brushless DC motors are standard for most quad copters due to their efficiency, reliability, and power-to-weight ratio. Motor specifications are typically described by their KV rating (RPM per volt), with lower KV motors generating more torque for larger propellers and higher KV motors providing higher speeds with smaller propellers. Four DJI 2212 920 KV motors were used for the mission.

3. Propellers

Propellers convert motor rotation into thrust. They come in various sizes and pitches, affecting efficiency, noise, and power consumption. Propellers must be balanced to prevent vibrations that can affect flight stability and sensor readings. Propellers were chosen according to motor rating and total thrust required for the mission which can be obtained from following data sheet.

ML2212 MOTOR								
Item NO.	Volts (V)	Prop	Throttle	Amps (A)	Watts (W)	Thrust (g)	Efficiency (g/W)	Operating temperature(℃)
ML2212 920KV	11.1V	DJI9. 4*4. 3	50%	1. 8	20. 0	230	11. 5	37℃
			65%	2. 8	31. 1	310	10. 0	
			75%	3. 9	43. 3	410	9. 5	
			85%	5. 5	61. 1	480	7. 9	
			100%	7. 6	84. 4	610	7. 2	
		APC10*4. 5	50%	2. 6	28. 9	290	10. 0	55℃
			65%	5. 1	56. 6	460	8. 1	
			75%	7. 4	82. 1	590	7. 2	
			85%	10. 1	112. 1	730	6. 5	
			100%	13. 4	148. 7	860	5. 8	
	14. 8V	DJI9. 4*4. 3	50%	2. 7	40. 0	350	8. 8	52℃
			65%	4. 4	65. 1	490	7. 5	
			75%	6. 3	93. 2	640	6. 9	
			85%	8. 3	122. 8	790	6. 4	
100%			11. 5	170. 2	990	5. 8		

Notes: The test condition of temperature is motor surface temperature in 100% throttle while the motor run 10 min.environment temperature 24℃

5. Electronic Speed Controllers (ESCs)

ESCs regulate the power delivered to each motor based on commands from the flight controller. They convert the DC power from the battery into the three-phase AC power needed by brushless motors and control motor speed precisely. Four ESCs of 30A each were used in our mission.

6. Flight Controller

The flight controller is brain of the quadcopter, processing sensor data and pilot inputs to control the motors. The Pixhawk 4 flight controller is used in our quadcopter which is an advanced flight controller that runs PX4 firmware, offering features like position holding, return-to-home, and autonomous mission execution.

7. Power Distribution Board (PDB)

It distributes power from the battery to the ESCs and other electronic components, often incorporating voltage regulators to provide appropriate power levels for different components.

8. Battery

Lithium Polymer (LiPo) batteries are most common due to their high energy density and discharge rates. Battery capacity (measured in mAh) and discharge rate (C rating) significantly impact flight time and performance. HRB 4500 mAh 60 C li-poly battery was used.

9. Radio Control System

It consists of a transmitter and a receiver. Modern systems use digital protocols operating in the 2.4GHz or 900MHz bands, with features like frequency hopping for reliability. Futaba T6K 8 channel transmitter/receiver was used for the mission.

10. Sensors

Modern flight controllers like the Pixhawk 4 incorporate various sensors:

- Accelerometer: Measures linear acceleration
- Gyroscope: Measures angular velocity
- Magnetometer (compass): Determines orientation relative to Earth's magnetic field
- GPS: Provides global position information. M8N GPS module was used for our mission.

Accelerometer, Gyroscope and Magnetometer are part of Pixhawk 4 onboard sensor suite whereas GPS is connected externally to Pixhawk.

11. Servo Mechanism:

A servo motor controlled by the Pixhawk 4 operates the release mechanism. Standard MG90S servo was used for dropping due to their reliability and precise control.

Assembling the quadcopter

- Attach the F450 frame parts securely using screws.
- Mount the motors onto the designated arms of the frame.
- Install the ESCs (Electronic Speed Controllers) near each motor and connect them.
- Secure the power module and connect it to the battery and flight controller.
- Attach the Pixhawk 4 flight controller to the center of the frame.
- Connect the GPS module to the flight controller.
- Mount the transmitter/receiver for remote control functionality.
- Connect the telemetry system for data monitoring.
- Attach the propellers to the motors (ensure correct orientation).
- Secure the LiPo battery and connect it to the power module.
- Connect external servo to flight controller.
- Calibrate the flight controller and check all connections.

Software setup for Autonomous Mission

QGroundControl

QGroundControl (QGC) is an open-source ground control station (GCS) application designed for managing and operating drones and other unmanned vehicles. Its user-friendly interface and support for various autopilots (e.g., PX4, ArduPilot) make it a popular choice for drone operators and developers.

Initial Vehicle Setup with PX4

Setting up a Pixhawk 4 with PX4 firmware involves these essential steps:

1. Installing QGroundControl:

- Download and install QGC on computer.

2. Connecting the Pixhawk:

- Connect the Pixhawk 4 to the computer via USB.

3. Firmware Installation:

- Open QGC and navigate to Vehicle Setup.
- Select Firmware.
- Choose the PX4 firmware version appropriate for application.
- Follow the on-screen instructions to flash the firmware.

4. Airframe Configuration:

- Select the appropriate airframe type (S500 generic).
- Apply and restart the controller.

5. Sensor Calibration:

- Calibrate the compass following the on-screen rotation instructions.
- Calibrate the gyroscope by keeping the vehicle still.
- Calibrate the accelerometer by positioning the vehicle in various orientations.
- Calibrate the airspeed sensor if equipped.
- Set up the level horizon.

6. Radio Setup:

- Bind your transmitter and receiver according to manufacturer instructions.
- Connect the receiver to the Pixhawk.
- Use QGC to calibrate the control channels.
- Configure and test failsafe settings.

7. Flight Modes:

- Assign flight modes such as stabilized, mission, manual, RTL etc. to transmitter switches.
- Configure parameters for each flight mode.
- Test mode switching.

8. Power Setup:

- Configure battery parameters (cells, capacity, warning thresholds).
- Calibrate voltage and current sensors if equipped.

9. Failsafe modes:

- Navigate to Vehicle Setup > Safety.
- Configure and set up all the failsafe modes and action to be taken.

10. Servo Configuration in:

- Navigate to Vehicle Setup > Parameters
- Find the SERVO_AUX_FUNCTION parameters.
- Set an unused servo output (e.g., SERVOx_FUNCTION) to Manual (value: 0) or another appropriate function.

11. PWM Range Configuration:

- Set SERVOx_MIN and SERVOx_MAX to define the servo's range of motion.
- Typical values are 1000 for minimum and 2000 for maximum.
- Adjust these values based on your specific servo and mechanism

12. Control Channel Assignment:

- For manual control, assign a transmitter channel to the servo function.
- Navigate to Radio setup and ensure the channel is properly calibrated.
- Test the servo movement using the transmitter.

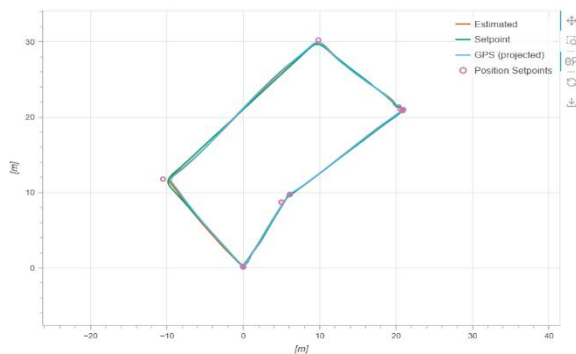
13. Mission Integration:

- Open the Mission Planning view in QGC interface.
- Create a mission with waypoints including takeoff.
- Add a "DO_SET_SERVO" command at the appropriate waypoint.
- Specify the servo number and PWM value to trigger the release mechanism.
- Insert Geofencing (if needed) for safe operation.
- Upload the mission to pixhawk.

14. Testing the Mechanism:

- Set COM_PREAMP_MODE to always.
- Test the servo operation while the quadcopter is disarmed.
- Verify that the mechanism releases smoothly and reliably.
- Conduct ground tests with the payload attached.

Results



Conclusion

Autonomous dropping using a Pixhawk 4-equipped quadcopter represents a sophisticated application of UAV technology with numerous practical applications. By understanding the fundamental components and working principles of quadcopters, properly configuring the hardware and software through QGroundControl, and carefully testing the system, operators can implement reliable autonomous dropping capabilities.

The continued development of flight control software like PX4, along with improvements in sensor technology and power systems, will further enhance the precision, reliability, and safety of autonomous dropping operations. As regulatory frameworks evolve to accommodate advanced UAV operations, we can expect to see broader adoption of this technology across various industries.